

Edi Zhang

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[Website](#) | [Linkedin](#) | [Github](#)

Education

University of California, Los Angeles (UCLA) | Los Angeles, CA

Sep 2023 - Jun 2027

BS, Applied Mathematics, Minor in Data Science Engineering, Specialization in Computing

- **GPA:** 3.85/4.0 — Dean's Honors List; ACT: 35
- **Core Math/EECS Coursework:** Real Analysis I/II, Dynamical Systems, Discrete Math, Differential Equations, Data Structures/Algorithms, Classical/Quantum Mechanics, Systems and Signals, Mathematical Modeling;
- **Machine Learning Coursework:** Neural Networks/Deep Learning I/II (Graduate), Applied Numerical Computing, Abstract Linear Algebra (Honors), Computational Imaging (Graduate), Probability, Statistics;
- **Skills:** Python, C++, MATLAB, Data Analysis (NumPy, Pandas, Matplotlib), Machine Learning (PyTorch), SQL, Julia, ReactJS (HTML/CSS/JS), Excel, Indesign

Work Experience

Capital One | Software Engineer Intern | Richmond, VA

Jun 2025 - Present

- Developing metric summaries to visualize compliance vulnerabilities across internal systems using React and Next.js
- Pulling data from external sources into a centralized platform using Glue ETL, improving developer efficiency by 25%
- Transformed manual data pulling into automated data workflows by creating DAGs using Apache Airflow and PySpark for scheduled batch processing of vulnerability reports, allowing for displayed data summaries to stay up-to-date
- Designing APIs using PostGraphile and GraphQL to fetch Qualys vulnerability scans from Snowflake database

Cleveland Clinic Lerner Research Institute | Researcher | Cleveland, OH

Jun 2024 - Mar 2025

- Researched in the [Advanced Musculoskeletal Imaging Lab](#) on mathematical modeling of tibiofemoral joint using Generalized Procrustes and Principal Component statistical analyses techniques to analyze directions of bone shape variance
- Constructed topological surfaces by processing 275 MRI masks using Marching Cubes and Laplacian Smoothing
- Implemented variance decomposition on five surface models to obtain hyper-dimensional eigenvectors representing principal directions of shape variance to ultimately improve characterization of knee osteoarthritis development
- Evaluated model by reconstruction accuracy, compactness, osteoarthritis severity stratification, and variance directions
- Presented [abstract](#) at 2025 Orthopaedic Research Society Annual Conference and preparing manuscript for publishing

UCLA Department of Mathematics | Course Reader | Los Angeles, CA

Sep 2024 - June 2025

- Generating XML files and Python scripts to efficiently score homework assignments for C++ programming course
- Grading assignments for over 90 students, ensuring timely feedback and detailed evaluations for coding projects
- Assisting TAs/LAs with clarifying programming concepts and offering support on debugging and optimizing code

Extracurricular Activities

The Bruin Group Consulting | Senior Consultant | Los Angeles, CA

Jan 2024 - Present

- Engaging in pro-bono strategy/marketing projects through premier consultancy (4.5% acceptance rate) for startups
- Spearheaded market research on M&A strategies for a multi-billion dollar top-5 U.S. aerospace and defense company
- Collaborated with vegan snack startup to research market entry analysis and pricing strategies towards college students

Bruin Sport Analytics | Data Journalist | Los Angeles, CA

Jan 2024 - Jun 2024

- Leveraged in-depth data analysis and statistical techniques to publish data-driven articles on sports performances
- Presented and pitched article ideas and data processing/analysis methods at workshops for data science fundamentals
- Underwent 10-week statistics/data science course with NumPy, Pandas, Tensorflow, Matplotlib, Sklearn, PyTorch

Projects

EMG Classification | *Python, PyTorch* | [Github](#)

Feb 2025 - Mar 2025

- Tested EMG Encoder to measure keystroke classification accuracy using various architectures including CNNs (75.41%), LSTMs (77.52%), GRUs (77.05%), CNN + LSTM (79.47%), and CNN + GRU (80.16%) on emg2qwertz dataset
- Achieved 80.55% accuracy by fine-tuning using CNN + GRU + Gaussian Noise and analyzed other combinations of fine-tuning methods including Frequency masks, Downsampling, and Random Cropping to optimize performance
- Analyzed test and validation accuracy and model compactness across architectures to optimize generalizability

PyTorch Implementation | *Python, NumPy*

Jan 2025 - Feb 2025

- Built FFNNs and CNNs from ground-up and implemented functions and parameters such as backprop, batch-norm, k-nearest neighbors, k-means, softmax classifier, Adam/RMSProp/SGD optimizers, momentum, dropout using NumPy
- Optimized and tested these Neural Networks on CIFAR-10 to achieve 75% validation accuracy on image classification

Temple of Doom | *C++* | [Github](#)

May 2024 - Jun 2024

- Multiple-level single player game managing 50+ actors and objects each player move using dynamic memory allocation
- Uses polymorphism and inheritance to create 15 types of actors/objects which manage game interactions each move

Additional Information

Competitions: BCG Case Competition Finalist, ACM Hackathon, ACM ICPC, ASA Datafest, Integration Bee, ICCA

Achievements: AIME Qualifier (2-time), AP Scholar with Distinction, National Merit Commended Scholar, Monte Vista Athletic Booster Scholarship, College Club Swimming Nationals Qualifier, USA Swimming CA/NV Sectionals Finalist

Volunteering: Campus Barber, Awaken A Cappella, D1 Swim Team Volunteer, UCLA Volunteer Day, C1 Tech Gives Back

Interests: Blitz Chess (2200), Swimming, Floral Designing, Barbering, Musical Theater, Speedcubing, Jazz Piano